SEMESTER - IV

COURSE: MAJOR – 5 (MJ-5) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: INDIAN KNOWLEDGE SYSTEM: PHYSICS IN ANCIENT INDIA **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

Analyze Vaisheshika, Nyaya, Saṃkhya, and Vedanta frameworks for physical concepts. Critically evaluate ancient Indian theories of matter, motion, cosmology, and mathematics. Examine epistemological methods (deduction, observation, yukti/reasoning) vs. Modern scientific methodology. Assess the transmission and global impact of Indian ideas.

Learning Outcomes:

1. Understand Vaisesika and Nyaya frameworks on atomism, motion, and causation in ancient Indian physics.
2. Analyze early Indian ideas on gravity, force, and alchemy, comparing them with Newtonian concepts.
3. Explore contributions of Indian astronomy and mathematics, including planetary models and trigonometric developments.
4. Identify parallels between Indian philosophical systems and modern physics theories.
5. Recognize the contributions of Indian scientists to classical and modern physics.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Foundations of Indian Physics (20 Lectures): Vaisesika theory of atomism: parmaṇu, categories of reality, padarthas (dravya, guṇa, karma, etc.), Nyaya integration: logic, inference, epistemology (gyan mimansa) applied to natural phenomena (e.g., force, motion), Detailed study: Kaṇada's sutras, classifications of motion and physical causation. Early ideas of motion, gravity (Gurutva), and alchemy (Rasavadam), Comparative framework versus Newtonian gravity. Indian astronomy: Surya Siddhanta, Siddhanta Jyotiṣa, planetary models by Aryabhaṭa and Nilakaṇṭha Somayaji, Mathematics: Decimal place value, geometry in Śulba Sutras, Kerala school's development of trigonometry and proto calculus, Solar worship, speed of light calculations in Vedic texts, Vedic acoustics.

UNIT-II:

IKS Ideas & Modern Physics (10 Lectures): Cross cultural resonance: Correspondences between Indian philosophical ideas (e.g. Vedanta, Samkhya etc.) and modern quantum & cosmological concepts, Consciousness and cosmology: Indian models and their relevance to theories of time, Cyclic universe, Mind matter interaction.

UNIT-III:

Indian Scientists and their contributions in Physics (15 Lectures): Aryabhatta: heliocentrism, planetary motion, circumference of Earth. Bhaskaracharya: calculus, gravity, planetary models. Acharya Kanada: atomic model, classification of matter. C.V. Raman and Raman Effect, S.N. Bose and Bose-Einstein Statistics, Homi Bhabha and Indian atomic energy program, Vikram Sarabhai and ISRO, Meghnad Saha and astrophysics, Subrahmanyam Chandrasekhar and black holes, Contribution of Vashishtha Narayan Singh in in the field of relativity.

SUGGESTED READINGS:

1. Introduction to Indian Knowledge System: Concepts & Applications by B. Mahadevan, Vinayak B Rajat and Nagendra Pavana, PHI Learning Pvt Ltd. Delhi.
2. Indian Knowledge System by Sunanda M. Kangane and N. S. Jadhav,
3. Theoretical Physics and Indian Philosophy: Conceptual Coherence (Sidorova Biryukova)
4. Nyaya Vaiseṣika conceptual treatises (e.g. Narayan's overview)
5. Vaiseṣika Sutra of Kaṇāda (English trans., e.g. by Nandalal Sinha)

6. R.N. Rai (2014), Indian Physics: Outline of Early History

7. Kim Plofker (2009), Mathematics in India.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Demonstrate sound vibration using tuning forks, analyze pitch, amplitude; discuss Nada-Brahma.
2. Demonstrate a structured Nyaya-style inference (anumana) for a physical event.
3. Construct a basic Panchanga-based calendar noting tithis, nakshatras, and planetary conjunctions.
4. Analyze the famous Rigveda verse (by Sayana) estimating speed of light; compare numerically with modern values.
5. Construct altars using Pythagorean triples as described in Śulba Sūtras.
6. Predict sunrise/sunset or planet positions using simplified algorithms from Surya Siddhanta.
7. Derive sine and cosine values using Madhava's series; compare with modern Taylor expansions.
8. Construct a simple ancient Indian timekeeping device using water flow.

SUGGESTED READINGS:

1. Introduction to Indian Knowledge System: Concepts & Applications by B. Mahadevan, Vinayak B Rajat and Nagendra Pavana, PHI Learning Pvt Ltd. Delhi.

2. Indian Knowledge System by Sunanda M. Kangane and N. S. Jadhav,

3. Theoretical Physics and Indian Philosophy: Conceptual Coherence (Sidorova Biryukova)

4. NyayaVaiseṣika conceptual treatises (e.g. Narayan's overview)

5. Vaiseṣika Sutra of Kaṇāda (English trans., e.g. by Nandalal Sinha)

6. R.N. Rai (2014), Indian Physics: Outline of Early History

7. Kim Plofker (2009), Mathematics in India.

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SEMESTER - IV

COURSE: MAJOR – 6(MJ-6) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: HEAT AND THERMODYNAMICS **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to provide students with a thorough understanding of the laws of thermodynamics, kinetic theory of gases, quantum theory of radiation, and statistical mechanics, enabling them to analyze and solve complex thermodynamic problems and phenomena.

Learning Outcomes:

1. Understand the fundamental principles of thermodynamics, including the zeroth, first, second, and third laws, and apply them to analyze heat and work interactions, thermodynamic processes, and entropy changes.
2. Describe the kinetic theory of gases, including Maxwell-Boltzmann distribution, mean speeds, degrees of freedom, and the law of equipartition of energy, and apply it to calculate specific heats of gases.
3. Explain the quantum theory of radiation, including Planck's law of blackbody radiation and its implications for spectral distribution, energy density, and various radiation laws.
4. Analyze statistical mechanics concepts such as macrostates, microstates, entropy, and thermodynamic probability, and apply distribution laws like Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac to describe the behavior of gases at different temperature regimes.
5. Apply theoretical principles from thermodynamics, kinetic theory, quantum theory, and statistical mechanics to solve problems related to heat, work, radiation, and gas behavior in various physical systems.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Fundamentals of Gases and Thermodynamic Variables (10 Lectures): Kinetic Theory of Gases: Maxwell-Boltzmann Law of Distribution of Velocities, Degrees of Freedom, Law of Equipartition of Energy, Specific Heats of Gases. Real Gases and Thermodynamic Properties: Behavior of Real Gases, Virial Equation, Critical Constants and Boyle Temperature, Vander-Waal's Equation of State.

UNIT-II:

Laws of Thermodynamics and Applications (20 Lectures): Zeroth and First Law of Thermodynamics, Extensive and Intensive Thermodynamic Variables, Thermodynamic Equilibrium, Concept of Work and Heat, State Functions, Internal Energy, General Relation between C_P and C_V, Work Done during Isothermal and Adiabatic Processes, Compressibility and Expansion Coefficient. Reversible and Irreversible Processes, Conversion of Work into Heat and Heat into Work, Heat Engines, Carnot's Cycle and Efficiency, Refrigerators and Coefficient of Performance, Kelvin-Planck and Clausius Statements, Carnot's Theorem, Clausius Inequality, Applications of Second Law, Entropy and Clausius Theorem, Entropy of a perfect gas. Principle of increase of Entropy, Entropy Changes in Reversible and Irreversible processes with examples, Mixing of entropy of two ideal gases, Entropy of the Universe, Temperature-Entropy Diagrams for Carnot's Cycle, Third Law of Thermodynamics, Unattainability of Absolute Zero.

UNIT-III:

Thermodynamic Potentials and Transport Phenomena (15 Lectures): Thermodynamic Potentials: Internal Energy, Enthalpy, Helmholtz Free Energy, Gibbs Free Energy. First and Second Order Phase Transitions: Clausius-Clapeyron Equation and Ehrenfest Equations. Derivations and Applications of Maxwell's Relations. Maxwell's Relations: Clausius-Clapeyron Equation, Values of C_p-C_v, T-ds Equations, Joule-Kelvin Coefficient for Ideal and Van der Waal Gases. Molecular Collisions and Transport Phenomena: Mean Free Path, Collision Probability, Viscosity, Thermal Conductivity, and Diffusion in Ideal Gases, Brownian Motion and its Significance. Thermal Conductivity: Rectilinear Flow of Heat in Metal Rod, Conductivity by Periodic Flow Method.

SUGGESTED READINGS:

1. Thermal Physics, S. Garg, R. Bansal and C. Ghosh, 1993, Tata McGraw-Hill.
2. A Treatise on Heat, Meghnad Saha, and B.N. Srivastava, 1969, Indian Press.
3. Thermodynamics, Kinetic theory & Statistical thermodynamics, F.W. Sears and G.L. Salinger. 1988, Narosa

4. University Physics, Ronald Lane Reese, 2003, Thomson Brooks/Cole.

5. Statistical Mechanics, R.K. Pathria, Butterworth Heinemann: 2nd Ed., 1996, Oxford University Press.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Determine the Coefficient of Thermal Conductivity of Cu by Searle's Apparatus.
2. Determine the Coefficient of Thermal Conductivity of a bad conductor by Lee and Charlton's disc method.
3. Study the variation of Thermo-emf of a Thermocouple with the Difference of Temperature of its Two Junctions.
4. Calibrate a thermocouple and determination of unknown temperature.
5. Determine of the ratio of the Specific Heats of a Gas by Clement and Desorme's apparatus.
6. Determine of temperature co-efficient of the resistance of the material of a wire.

SUGGESTED READINGS:

1. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
2. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, KitabMahal
3. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
4. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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SEMESTER - IV

COURSE: MAJOR – 7 (MJ-7) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: SEMICONDUCTOR DEVICES **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objectives

This course provides a fundamental understanding of semiconductor materials and devices, including diodes, transistors, and optoelectronic components. Students will learn the physical principles governing device behavior, analyze their characteristics and applications, and develop the ability to evaluate circuit performance in practical electronics and communication systems.

Learning Outcomes:

1. Understand the properties of semiconductor materials, band structures, and carrier dynamics.
2. Analyze p-n junction behavior under forward and reverse bias and apply diodes in rectification circuits.
3. Evaluate transistor operation in different configurations and understand load line and Q-point analysis.
4. Describe the working and characteristics of JFETs and MOSFETs in electronic circuits.
5. Explain the operation and applications of optoelectronic devices like LEDs, photodiodes, and solar cells.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Semiconductor Fundamentals (15 Lectures): Semiconductor Materials, Types of semiconductors, Charge carriers, Intrinsic and extrinsic materials. Direct and indirect bandgap semiconductors heavily doped semiconductors Carrier concentration, dependence of carrier concentration, and mobility, Electron and hole concentration equilibrium, Energy Bands: Introduction to energy bands, valence bands, conduction bands,

and band gaps, Density of states, Fermi level and dependence of Fermi level on the carrier concentration. Fermi-Dirac distribution.

UNIT-II:

p-n Junction Diodes and Applications (15 Lectures): P-N Junctions Formation, Depletion region, Forward and reverse bias, Diode characteristics, Junction Capacitance, barrier formation in p-n junction, barrier potential, forward and reverse biased diode, current flow mechanism in forward and reverse biased diodes, static and dynamic resistance. Diode as rectifier, load line and Q-point, half-wave rectifier, center-tapped and bridge full-wave rectifiers, calculation of average and RMS current and voltage, voltage regulation, ripple factor and rectification efficiency, filters. Zener Diode, Zener and avalanche breakdown, Zener diode as voltage regulator. Operation and characteristics of Light-emitting diodes (LEDs), Photodiodes, Solar cells, Semiconductor lasers.

UNIT-III:

Bipolar Junction Transistors (15 Lectures): Bipolar Junction Transistors (BJTs) Basic operation n-p-n and p-n-p Transistors, Characteristics of CB, CE, and CC Configurations, Current Gains and Relations between Them, Load Line Analysis of Transistors, DC Load Line and Q-point, Physical Mechanism of Current Flow, Active, Cutoff, and Saturation Regions, Field-Effect Transistors (FETs), Junction FETs (JFETs) Operation, characteristics, parameters. Metal-Oxide-Semiconductor FETs (MOSFETs): Ideal and real MOS capacitors, threshold voltage, output and transfer characteristics.

SUGGESTED READINGS:

1. Principles of Electronics, VK Mehta and Rohit Mehta, S Chand.
2. Integrated Electronics, J. Millman and C. C. Halkias, 1991, Tata Mc-Graw Hill.
3. Electronics: Fundamentals and Applications, J.D. Ryder, 2004, Prentice Hall.
4. Solid State Electronic Devices, B .G. Streetman & S. K. Banerjee, 6th Edn., 2009, PHI Learning.
1. Electronic Devices & circuits, S. Salivahanan & N. S. Kumar, 3rd Ed., 2012, Tata Mc-Graw Hill.
5. Electronic circuits: Handbook of design & applications, U. Tietze, C. Schenk, 2008, Springer.
6. Semiconductor Devices: Physics and Technology, S.M. Sze, 2nd Ed., 2002, Wiley India.
7. Electronic Devices, 7/e Thomas L. Floyd, 2008, Pearson India.
8. Hand book of electronics by Gupta and Kumar.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Study V-I characteristics of PN junction diode, and verification of diode equation.
2. Study the V-I characteristics of a Zener diode and its use as voltage regulator.
3. Estimate the energy gap of a semiconductor using a PN junction.
4. Study the characteristics of a Bipolar Junction Transistor in CE configuration.
5. Study the frequency response of voltage gain of a RC-coupled transistor amplifier.
6. Study the output and transfer characteristics of a JFET.
7. Design a common source JFET Amplifier and study its frequency response.

SUGGESTED READINGS:

1. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
2. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, Kitab Mahal
3. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
4. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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